

OSI Model - Technical Documentation

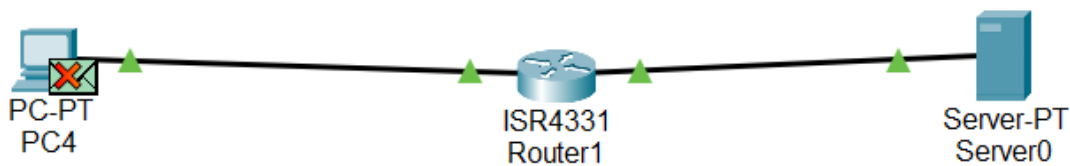
A Packet's journey: Encapsulation & Decapsulation across a network

Overview

This documentation explores the OSI model and the different layers within the OSI model, using packet tracer to better understand how data is sent through a network.

With this documentation, we used packet tracer, but this has a limitation where the Application, Presentation, and Session layers are not recorded and integrated within the software. I will provide this information at a conceptual level.

Network Topology



Device	Interface	IP address	Subnet Mask
P4	Fa0	192.168.1.2	255.255.255.0
Router 1	G0/0/0	192.168.1.1	255.255.255.0
Router 1	G0/0/1	192.168.2.1	255.255.255.0
Server	Fa0	192.168.2.2	255.255.255.0

Event List

Vis.	Time(sec)	Last Device
	450.451	--
	450.451	--
	450.452	PC4
	450.452	--
	450.453	PC4
	450.453	Router1
	450.454	Router1
	450.454	Server0
	450.455	Server0
	450.455	Router1
	450.456	Router1

Introduction

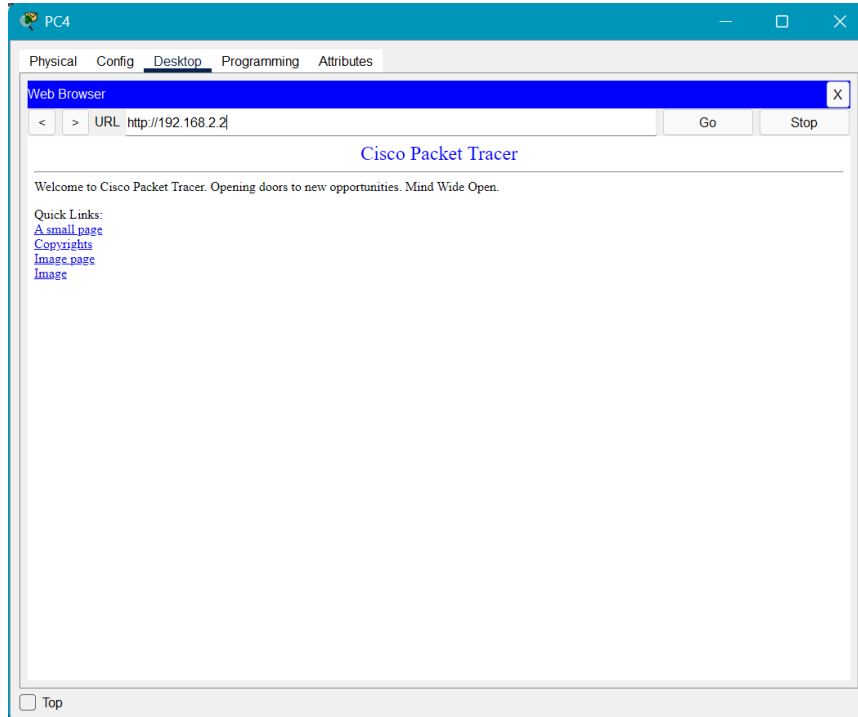
In the infancy of the Internet, different companies would use different type of proprietary hardware to ensure their devices would be able to talk to each other. Another way to imagine it, is having 6 people talking in different languages, that don't know how to talk to each other. So, to help with this, networking models were created and agreed how computers talk to each other on a higher level. The two models were: TCP/IP Model and the OSI Model.

Although many people use the TCP/IP model as their preferred model, the OSI model still is used to understand the foundational concepts of networking, and is the model we are using. The OSI model consists of the following layers:

Layer 7 - Application
Layer 6 - Presentation
Layer 5 - Session
Layer 4 - Transport Layer
Layer 3 - Network
Layer 2 - Data Link
Layer 1 - Physical

Application Layer:

Although packet tracer does not show this layer, we know that the user would be interacting with this layer, via the browser. Since the destination host is 192.168.2.2, we can use the IP address to access our server, just like a user is searching a web domain, such as youtube.



At this point, the PDU remains as data, and this data is passed onto the next layer, the presentation layer

Presentation Layer:

The Presentation Layer is responsible for formatting, encoding, and encrypting data so that the receiving device can understand it. Think of it as a translator — it ensures the data sent by PC4 is in a format that Server0 can read.

In this scenario, since we are using HTTP (not HTTPS), there is no encryption being applied. The data is transmitted as plain text, formatted in a way that both PC4's browser and Server0's web service can interpret. If we were using HTTPS, this is the layer where TLS encryption would be applied before the data travels down the stack. The PDU at this stage remains as data, and is passed down to the Session Layer.

Session Layer

In this layer, we are managing the session from the browser to the destination host, which in this case, is the server (192.168.2.2). The session layer is responsible for managing and terminating the session between the PC4 and Server0. So, when the server sees a new connection being made, the session layer is responsible for ensuring the connection remains open and does not close between PC4 and Server 0. When the host exits the website, the session layer comes in and ensures the connection between PC4 and Server 0 remains closed. We can think of this as the middleman between PC4 and Server 0, whose job is to ensure that there is a consistent stream of communication occurring and ending.

Transport Layer:

As we know, the Transport Layer is responsible for the reliability and flow-control of the data being sent over the medium. Looking at the PDU information below, we can see the TCP protocol has been used, along with some other information

TCP stands for Transmission Control Protocol, and protocol is a connection-oriented protocol. This means, before the data can be sent over the medium, a connection between the sender and the receiver has to be made. TCP uses a 3-way handshake, with the following process:

- SYN - This is telling the receiver that the host wishes to establish a connection with it
- SYN - ACK - This is telling the host, the server has received the SYN message
- ACK - The server is happy to establish a connection with the host.

PDU Information at Device: Server0	
OSI Model	Inbound PDU Details Outbound PDU Details
At Device: Server0 Source: PC4 Destination: HTTP CLIENT	
In Layers	Out Layers
Layer 7: HTTP	Layer 7: HTTP
Layer 6	Layer 6
Layer 5	Layer 5
Layer 4: TCP Src Port: 1026, Dst Port: 80	Layer 4: TCP Src Port: 80, Dst Port: 1026
Layer 3: IP Header Src. IP: 192.168.1.2, Dest. IP: 192.168.2.2	Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.2
Layer 2: Ethernet II Header 00E0.F97C.C402 >> 000A.F302.731E	Layer 2: Ethernet II Header 000A.F302.731E >> 00E0.F97C.C402
Layer 1: Port FastEthernet0	Layer 1: Port(s): FastEthernet0

Network Layer:

As we know, the Network layer is responsible for delivery of the packet over the network. Now, we ask: What does this mean? Let's say we have Bob (PC4) and Emily (Server0) living a few streets apart, and Bob wants to send an envelope to Emily. In this case, the envelope represents the data. The postman is the router. The postman job is to decide which route he must take to ensure he can reach Emily's house.

Now going back to the network, the router decides which route it must take to reach the destination. We can see this using the following command: "show ip int brief". This command displays the IP address on each interface.

```
Router(config)#do show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/0/1  192.168.2.1    YES manual up          up
GigabitEthernet0/0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router(config)#
```

So, we can see, to reach the destination 192.168.2.2, we have to use the int G0/0/1, since this is the closest matching IP address.

Data-Link Layer:

So, the data-link layer is responsible for hop-to-hop delivery. This means, when a segment reaches another device, on the way to reach the destination, MAC addresses are verified before opening the packet.

MAC addresses stand for Media Access Control, and these represent the actual device on a physical level rather than a logical level, that you would think on the network layer. Each MAC address is unique.

Before the router can decapsulate the packet, it must strip the segment, by removing the appropriate headers for that layer. But how does this happen? The router must verify the destination MAC address matches its own, and only then can the router reach the packet.

PDU Information at Device: Server0	
OSI Model Inbound PDU Details Outbound PDU Details	
At Device: Server0 Source: PC4 Destination: HTTP CLIENT	
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Layer 6	Layer 6
Layer 5	Layer 5
Layer 4: TCP Src Port: 1026, Dst Port: 80	Layer 4: TCP Src Port: 80, Dst Port: 1026
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Layer 2: Ethernet II Header 00E0.F97C.C402 >> 000A.F302.731E	Layer 2: Ethernet II Header 000A.F302.731E >> 00E0.F97C.C402
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Physical Layer:

The Physical Layer is the lowest layer of the OSI model and is responsible for the transmission of raw bits over the physical medium. At this point, all the headers from the layers above have been added, and the entire frame is converted into a stream of 1s and 0s and transmitted over the cable.

In this scenario, we can see from the PDU information that the physical interface being used is FastEthernet0. This means the bits are being transmitted electrically over a copper cable, at speeds of up to 100Mbps, from PC4 toward Router1.

It is important to note that the Physical Layer has no awareness of the data it is transmitting — it simply converts the frame into a signal and puts it on the wire. It does not care about IP addresses, MAC addresses, or port numbers. That logic lives in the layers above.